

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 13:

Friction

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Types of Friction

1. ***Dry Friction.*** Dry friction occurs when the unlubricated surfaces of two solids are in contact under a condition of sliding or a tendency to slide. A friction force tangent to the surfaces of contact occurs both during the interval leading up to impending slippage and while slippage takes place.

Types of Friction

2. *Fluid Friction.* Fluid friction occurs when adjacent layers in a fluid (liquid or gas) are moving at different velocities. This motion causes frictional forces between fluid elements, and these forces depend on the relative velocity between layers. When there is no relative velocity, there is no fluid friction. Fluid friction depends not only on the velocity gradients within the fluid but also on the viscosity of the fluid, which is a measure of its resistance to shearing action between fluid layers. Fluid friction is treated in the study of fluid mechanics and will not be discussed further here.

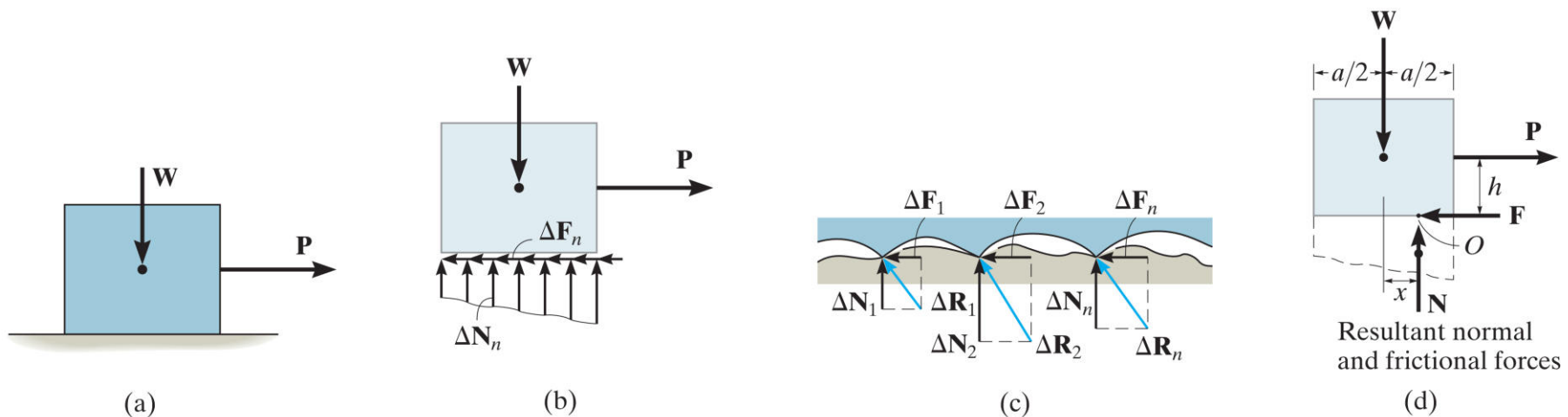
Types of Friction

3. ***Internal Friction.*** Internal friction occurs in all solid materials which are subjected to cyclical loading. For highly elastic materials the recovery from deformation occurs with very little loss of energy due to internal friction.

Characteristics of Dry (Coulomb) Friction

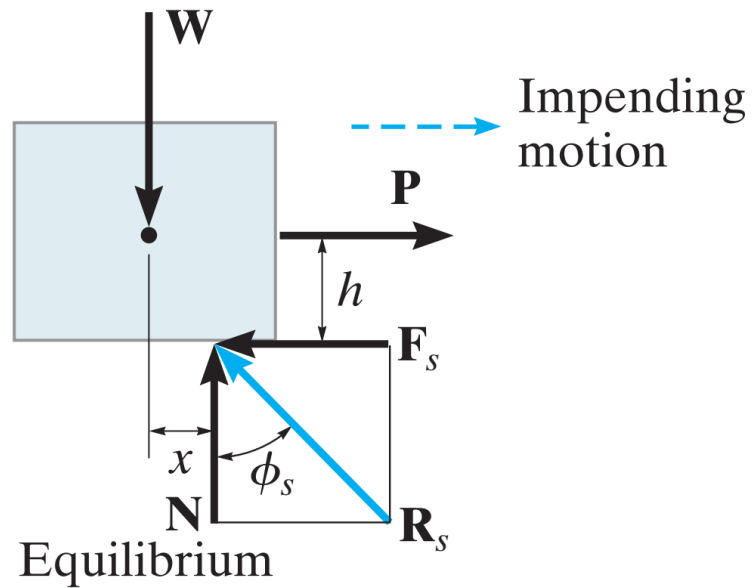
Friction is a force that resists the movement between two contacting surfaces that slide relative to one another. This force always acts *tangent* to the surface at the points of contact and is directed such to oppose the possible or existing motion between the surfaces.

Theory of Dry Friction.

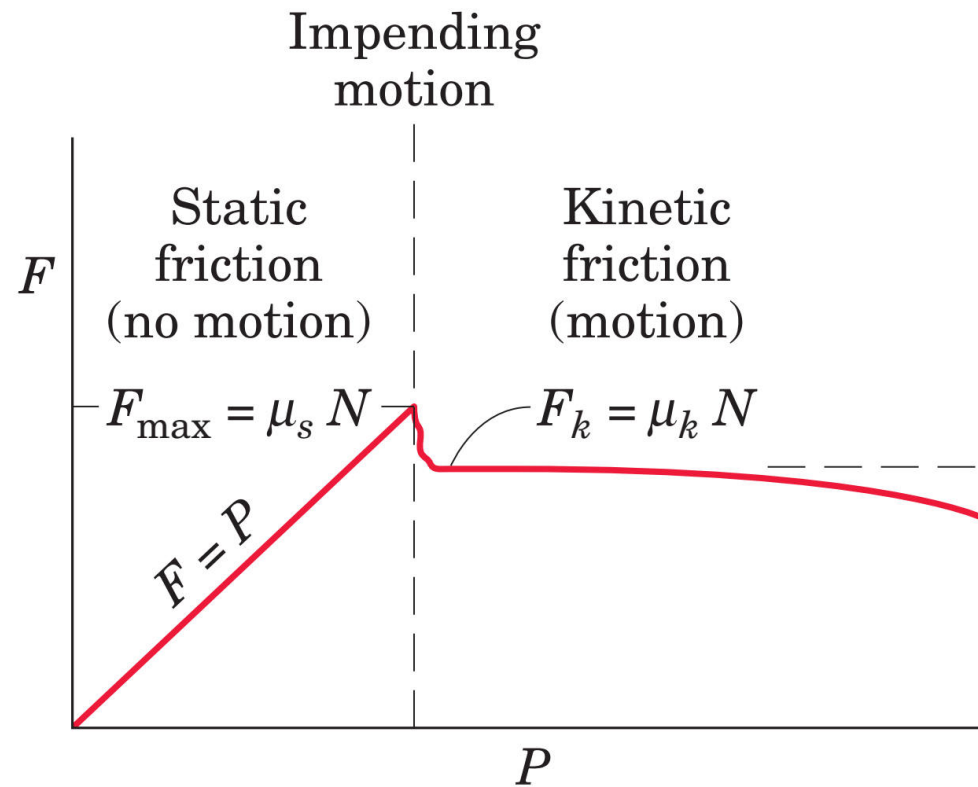


Characteristics of Dry (Coulomb) Friction

Equilibrium



Characteristics of Dry (Coulomb) Friction



Characteristics of Dry (Coulomb) Friction

Static Friction

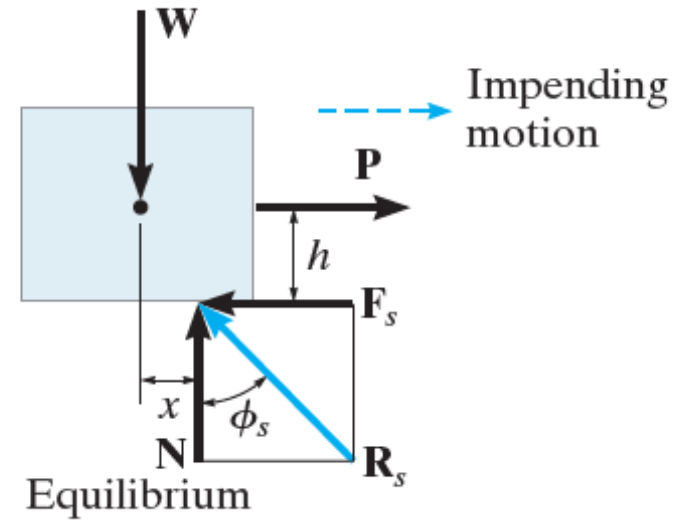
$$F < \mu_s N$$

Impending Motion Friction

$$F_s = \mu_s N$$

Kinetic Friction

$$F_k = \mu_k N$$



Characteristics of Dry (Coulomb) Friction

Contact Materials	Coefficient of Static Friction (μ_s)
Metal on ice	0.03–0.05
Wood on wood	0.30–0.70
Leather on wood	0.20–0.50
Leather on metal	0.30–0.60
Copper on copper	0.74–1.21

Characteristics of Dry (Coulomb) Friction

- F is a ***static frictional force*** if equilibrium is maintained.
- F is a ***limiting static frictional force*** F_s when it reaches a maximum value needed to maintain equilibrium.
- F is a ***kinetic frictional force*** F_k when sliding occurs at the contacting surface.
- The frictional force acts *tangent* to the contacting surfaces in a direction *opposed* to the *motion* or tendency for motion of one surface relative to another.

Characteristics of Dry (Coulomb) Friction

- The maximum static frictional force F_s that can be developed is independent of the area of contact between the surfaces, provided the normal pressure is not very low nor great enough to severely deform or crush the surfaces between the bodies.
- The maximum static frictional force is generally greater than the kinetic frictional force for any two surfaces of contact. However, if one of the bodies is moving with a *very low velocity* over the surface of another, F_k becomes approximately equal to F_s , i.e., $\mu_s \approx \mu_k$.
- When *slipping* at the surface of contact is *about to occur*, the maximum static frictional force is proportional to the normal force, such that $F_s = \mu_s N$.
- When *slipping* at the surface of contact is *occurring*, the kinetic frictional force is proportional to the normal force, such that $F_k = \mu_k N$.

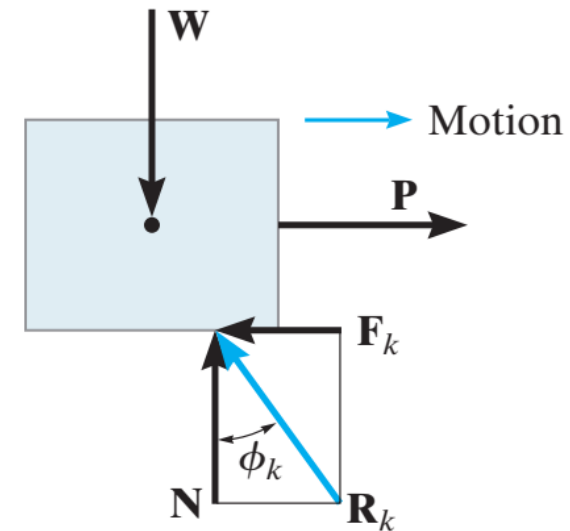
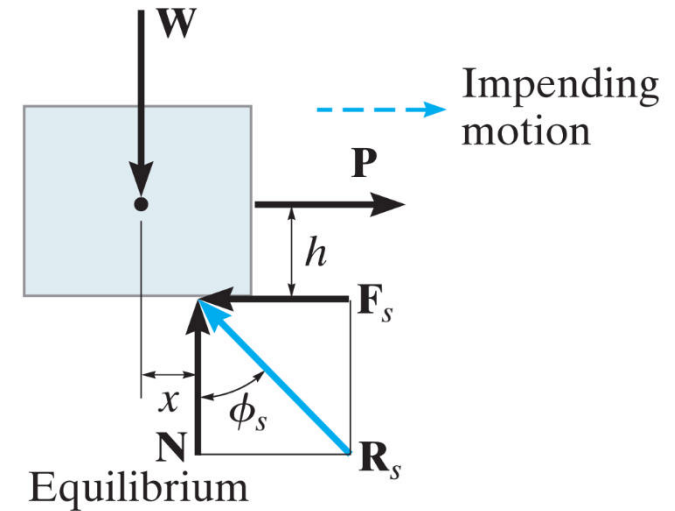
Angles of Friction

Angle of static friction:

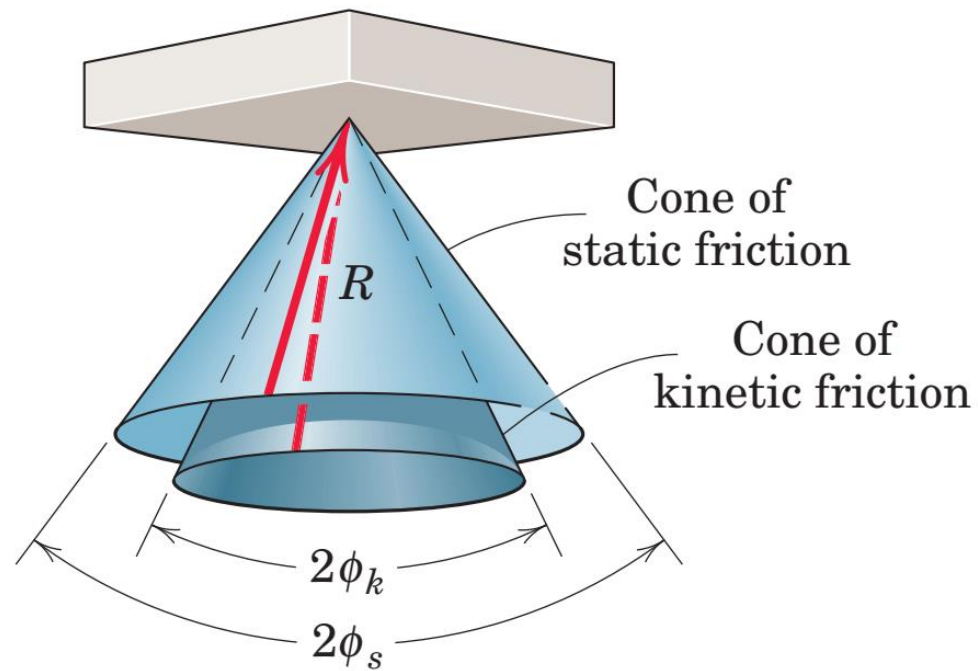
$$\phi_s = \tan^{-1}\left(\frac{F_s}{N}\right) = \tan^{-1}\left(\frac{\mu_s N}{N}\right) = \tan^{-1} \mu_s$$

Angle of kinetic friction:

$$\phi_k = \tan^{-1}\left(\frac{F_k}{N}\right) = \tan^{-1}\left(\frac{\mu_k N}{N}\right) = \tan^{-1} \mu_k$$

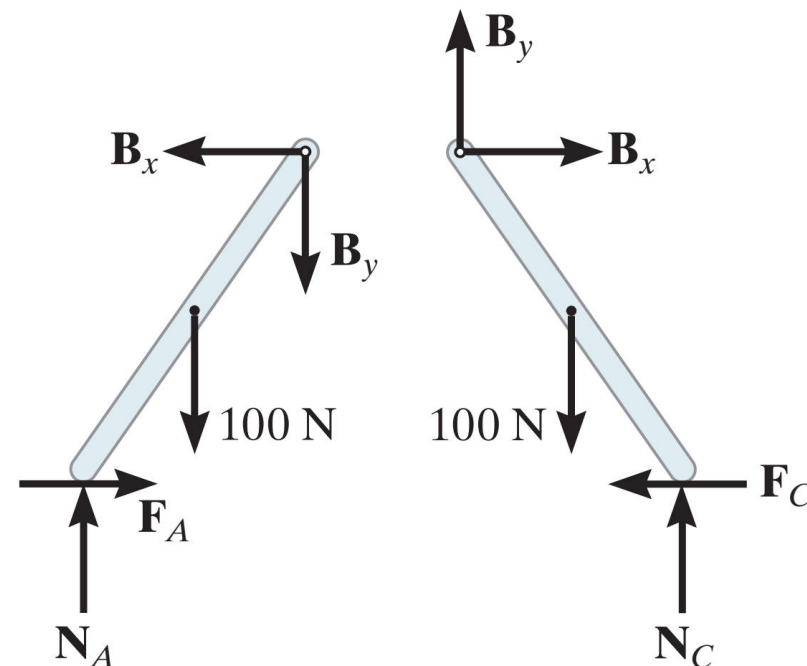
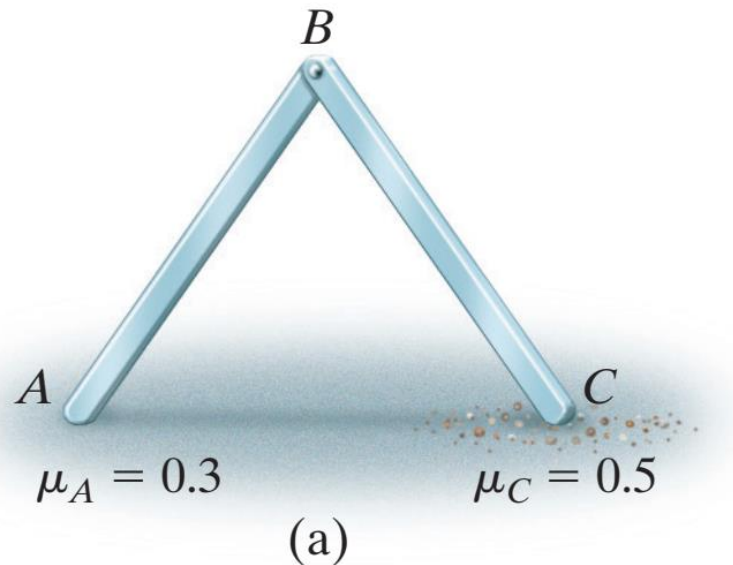


Angles of Friction



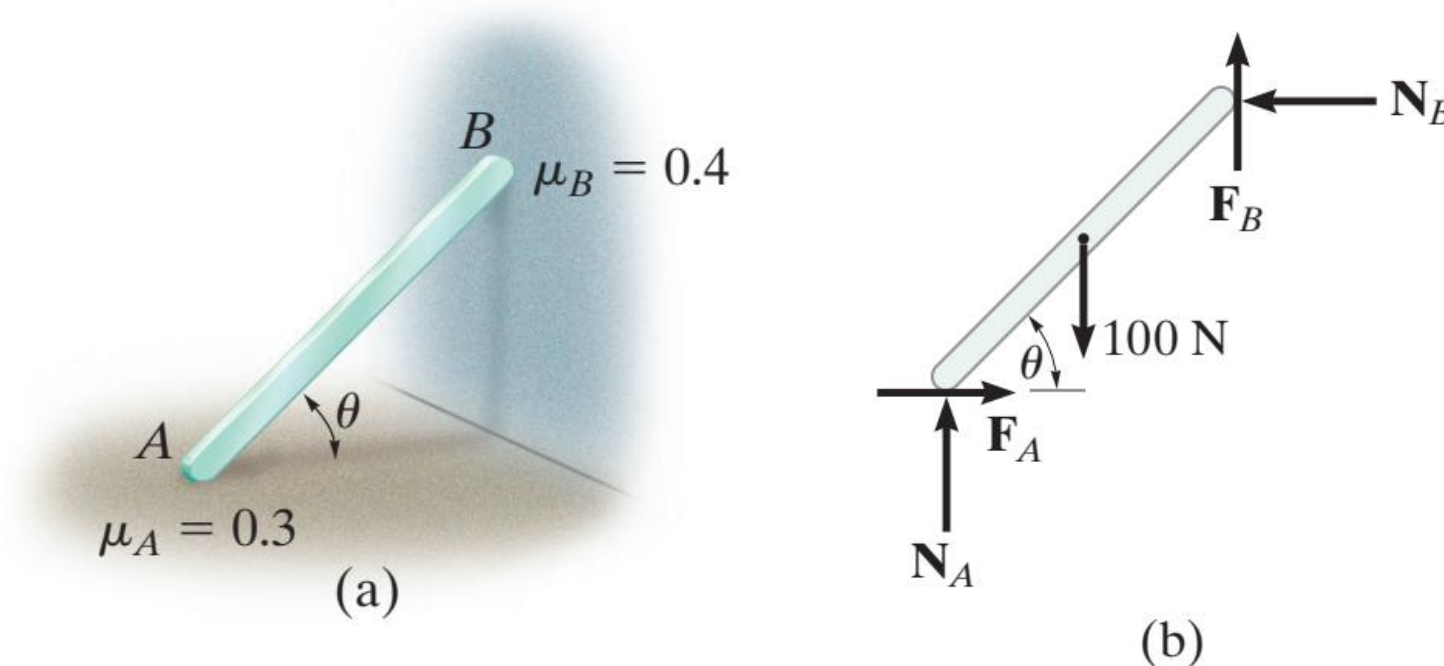
Dry Friction without Apparent Impending Motion

This requires the number of unknowns to be equal to the number of available equilibrium equations. Once the frictional forces are determined from the solution, however, their numerical values must be checked to be sure they satisfy the inequality $F \leq \mu_s N$; otherwise, slipping will occur and the body will not remain in equilibrium.



Dry Friction with Impending Motion at All Points of Contact.

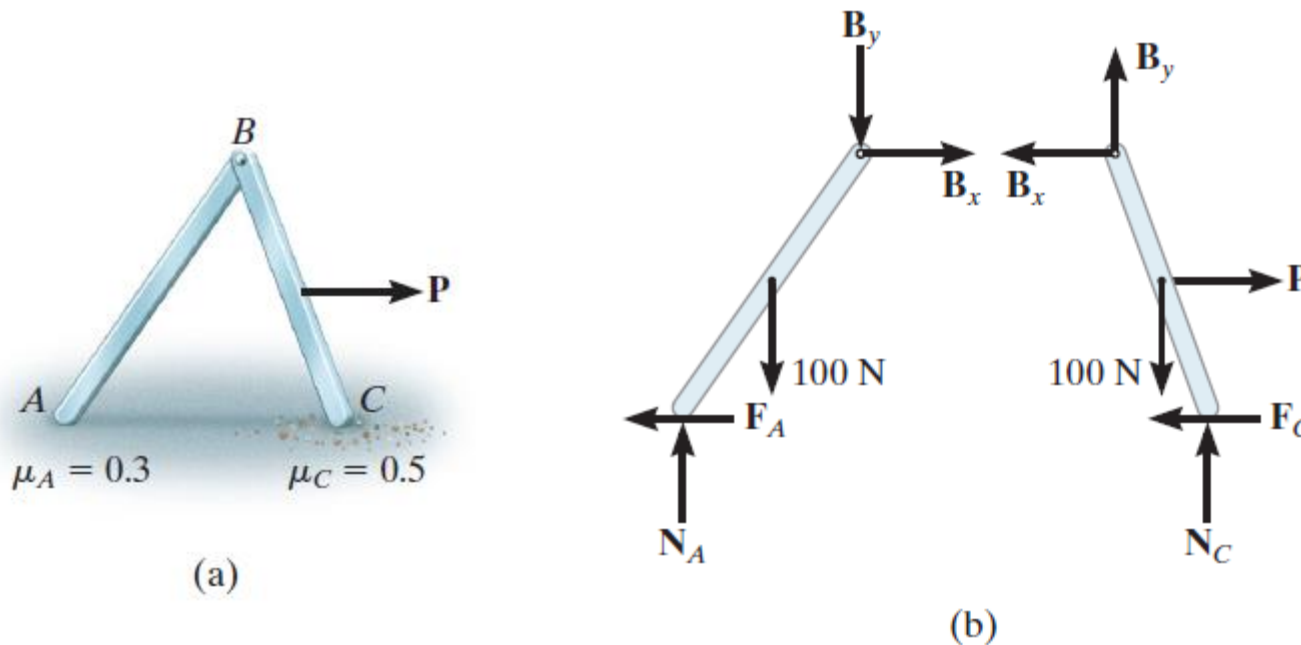
In this case the total number of unknowns will equal the total number of available equilibrium equations plus the total number of available frictional equations, $F = \mu_s N$.



finding the smallest angle θ at which the 100-N bar can be placed against the wall without slipping.

Dry Friction with Impending Motion at Some Points of Contact

For these types of problems, the number of unknowns will be less than the number of available equilibrium equations plus the number of available frictional equations or conditional equations for tipping.



Determining the horizontal force P needed to cause movement.

Concluding Notes

- Friction is a tangential force that resists the movement of one surface relative to another.
- If no sliding occurs, the maximum value for the friction force is equal to the product of the coefficient of static friction and the normal force at the surface, $F_s = \mu_s N$.
- If sliding occurs, then the friction force is the product of the coefficient of kinetic friction and the normal force at the surface, $F_k = \mu_k N$.
- There are three types of static friction problems. Each of these problems is analyzed by first drawing the necessary free-body diagrams, and then applying the equations of equilibrium, while satisfying either the conditions of friction or the possibility of tipping.

Example: The uniform crate shown below has a mass of 20 kg. If a force $P = 80 \text{ N}$ is applied to the crate, determine if it remains in equilibrium. The coefficient of static friction is 0.3.

$$\sum F_x = 0 \rightarrow 80 \cos 30^\circ - F_s = 0 \rightarrow F_s = 40\sqrt{3} \text{ N} = 69 < F_{s_{\max}}$$

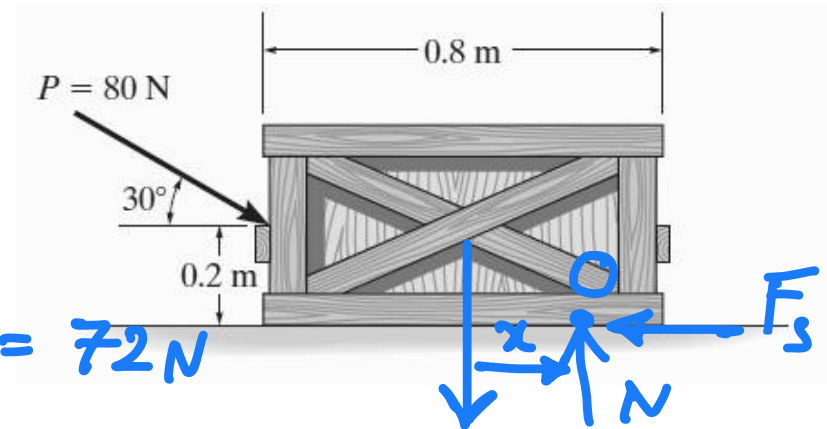
$$\sum F_y = 0 \rightarrow -80 \sin 30^\circ - 200 + N = 0$$

$$N = 240 \text{ N}$$

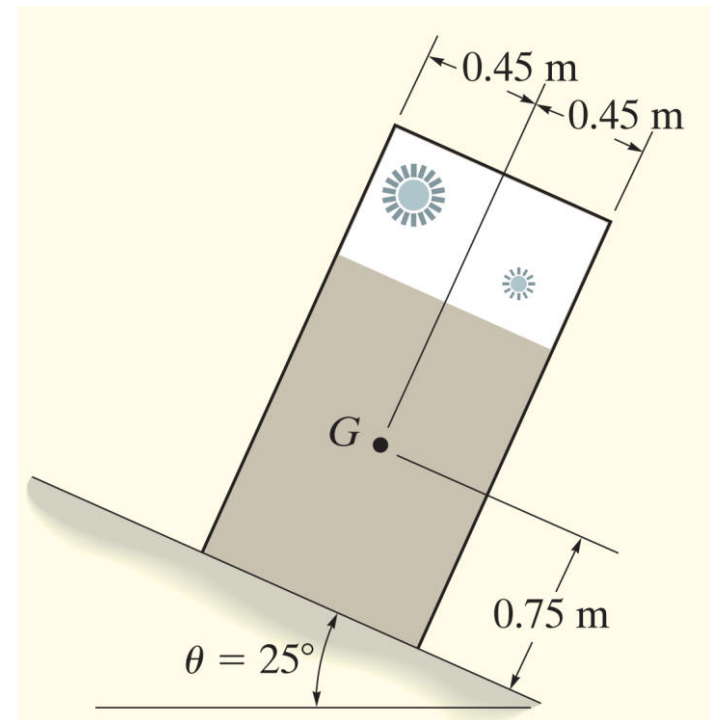
$$F_{s_{\max}} = \mu N = 0.3 \times 240 = 72 \text{ N}$$

$$\sum M_O = 0 \rightarrow -40\sqrt{3}(0.2) + 40(0.4 + x) + 200(x) = 0 \quad W = 200 \text{ N}$$

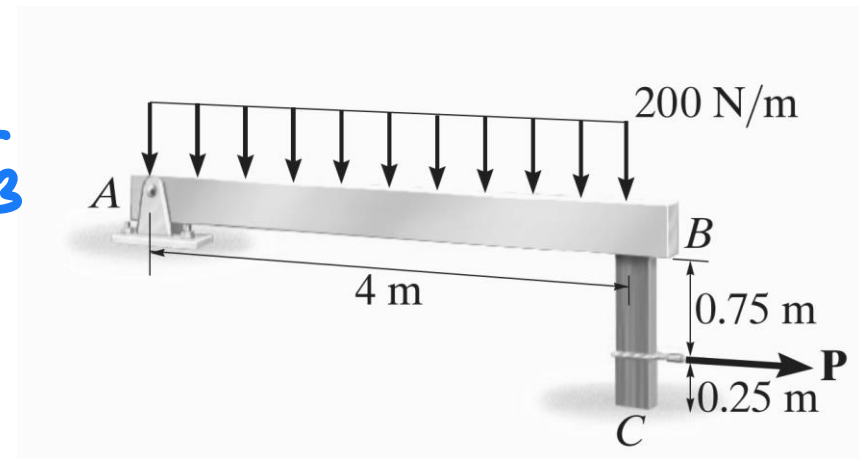
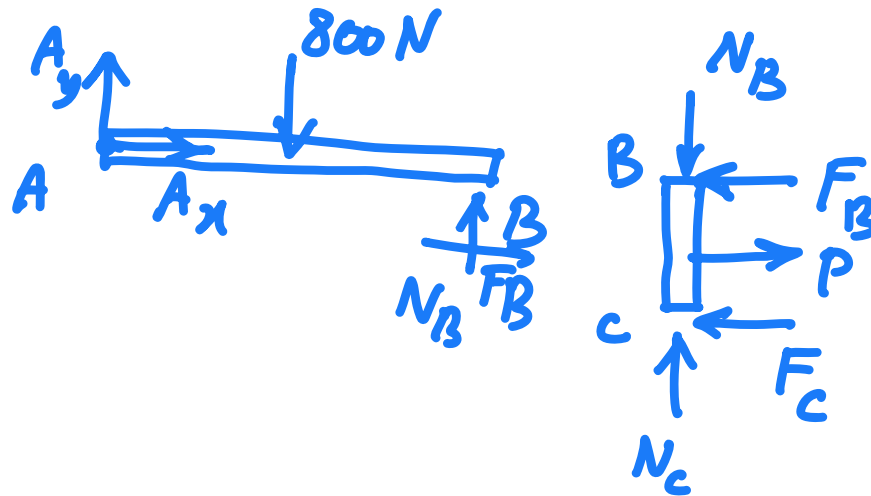
$$240x + 16 - 14 = 0 \rightarrow \left| x = \frac{-2}{240} \right| \checkmark < 0.4$$



Example: It is observed that when the bed of the dump truck is raised to an angle of $\theta = 25^\circ$ the vending machines will begin to slide off the bed. Determine the coefficient of static friction between a vending machine and the surface of the truck bed.

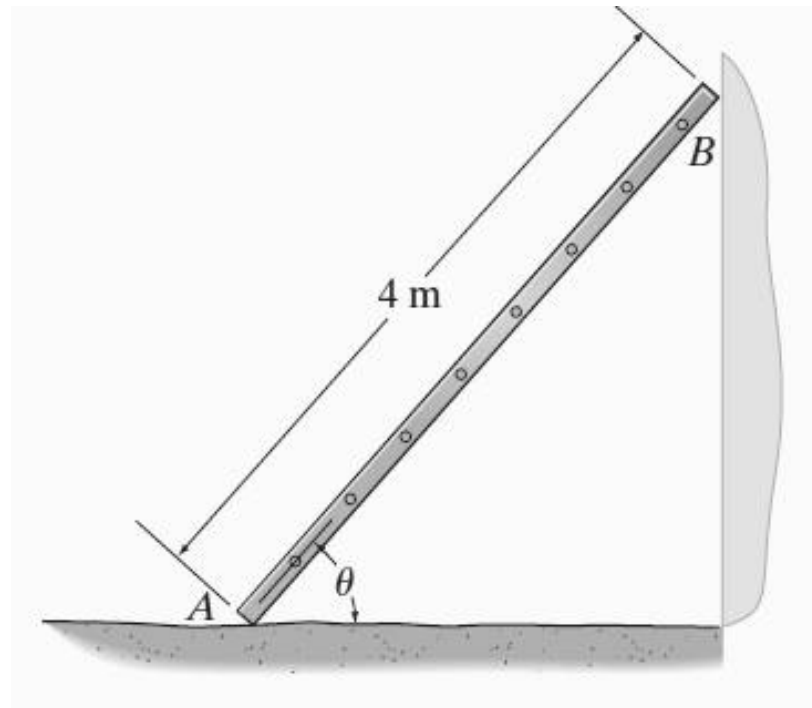


Example: Beam AB is subjected to a uniform load of 200 N/m and is supported at B by post BC . If the coefficients of static friction at B and C are 0.2 and 0.5 , determine the force $\mathbf{P}$ needed to pull the post out from under the beam. Neglect the weight of the members and the thickness of the beam.



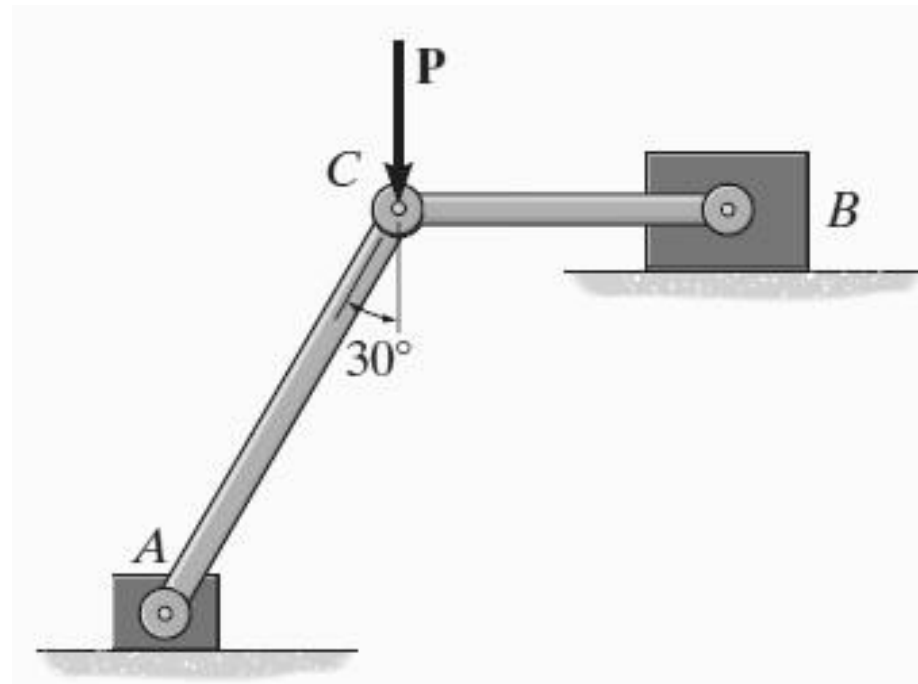
Practice at Home

1. The uniform 10-kg ladder rests against the smooth wall at B , and the end A rests on the rough horizontal plane for which the coefficient of static friction is 0.3. Determine the angle of inclination θ of the ladder and the normal reaction at B if the ladder is on the verge of slipping.



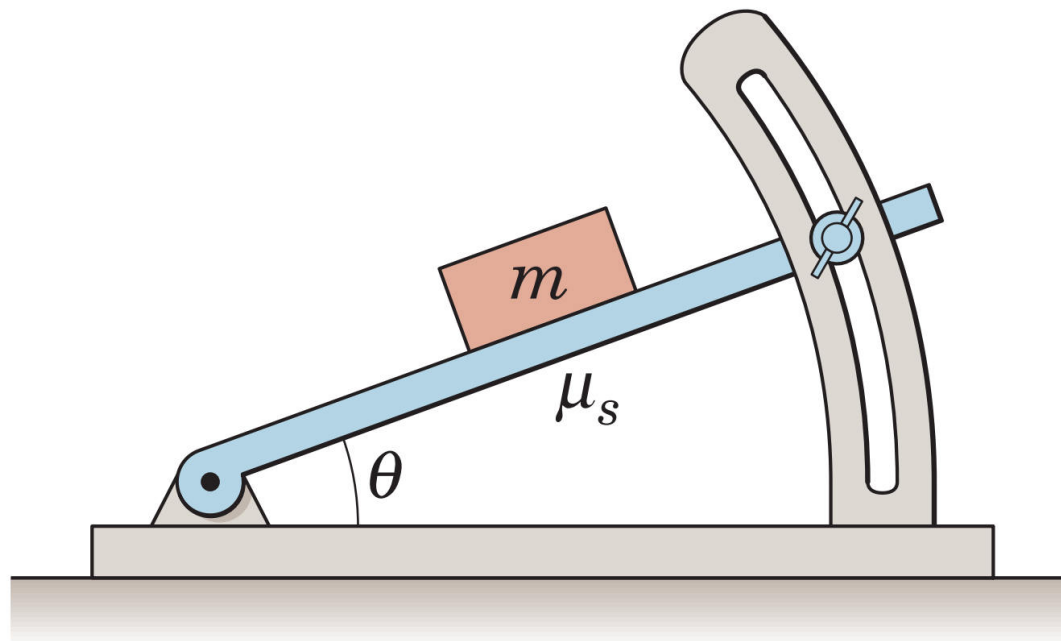
Practice at Home

2. Blocks *A* and *B* have a mass of 3 kg and 9 kg, respectively, and are connected to the weightless links. Determine the largest vertical force **P** that can be applied at pin *C* without causing any movement. The coefficient of static friction between the blocks and the contacting surfaces is $\mu_s = 0.3$.



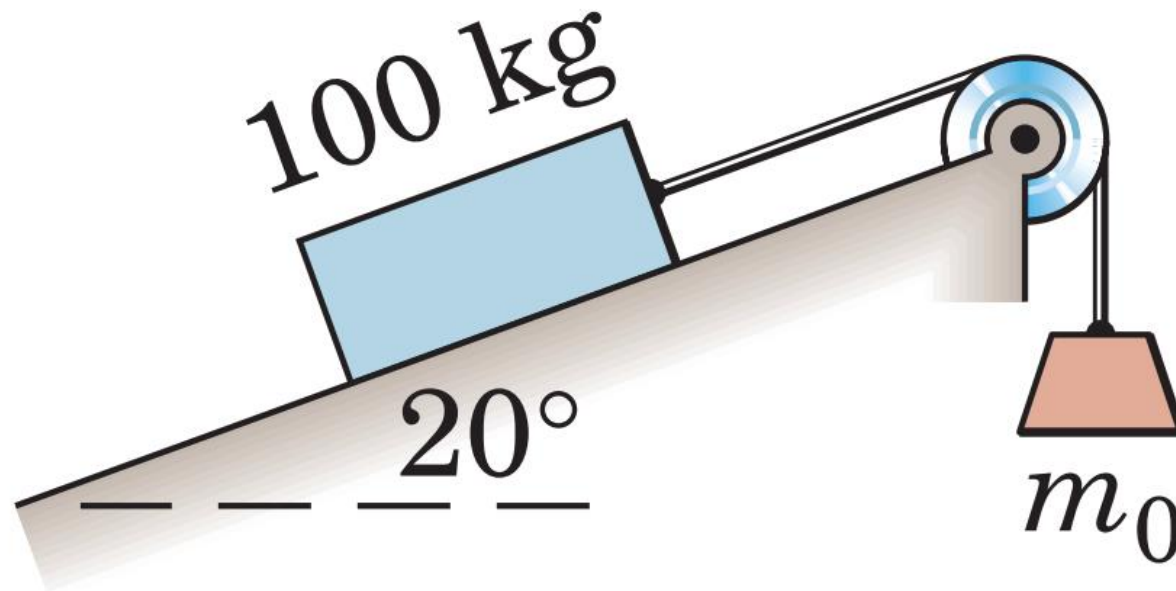
Practice at Home

3. Determine the maximum angle θ which the adjustable incline may have with the horizontal before the block of mass m begins to slip. The coefficient of static friction between the block and the inclined surface is μ_s .



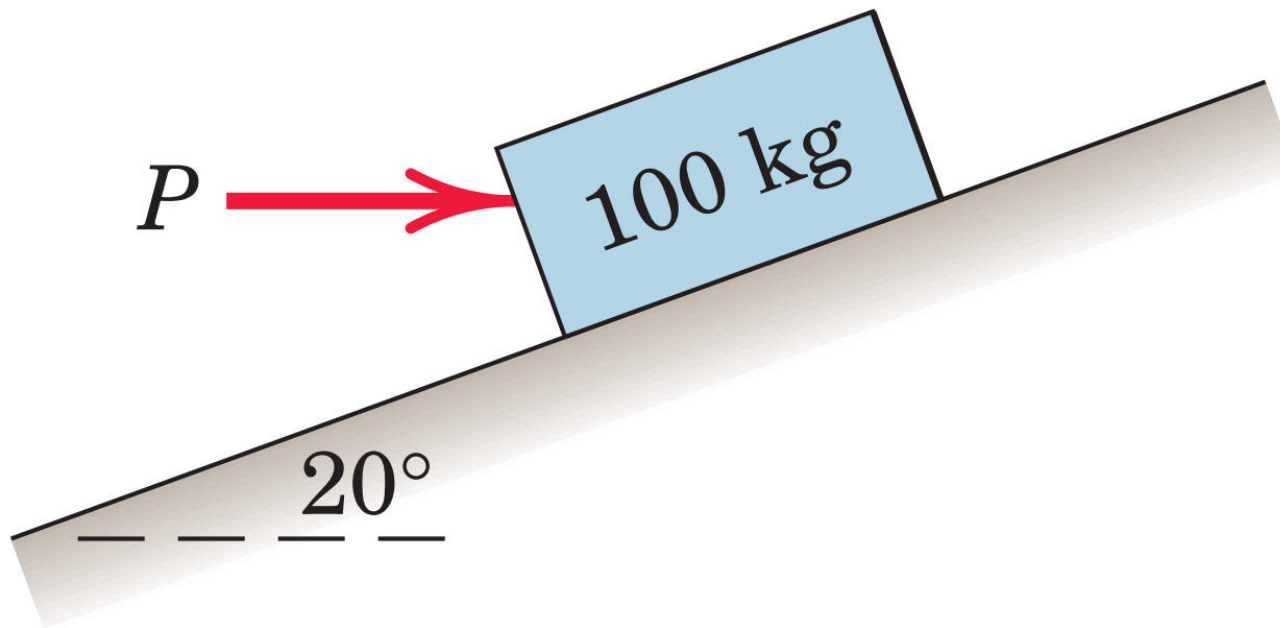
Practice at Home

4. Determine the range of values which the mass m_0 may have so that the 100-kg block shown in the figure will neither start moving up the plane nor slip down the plane. The coefficient of static friction for the contact surfaces is 0.30.



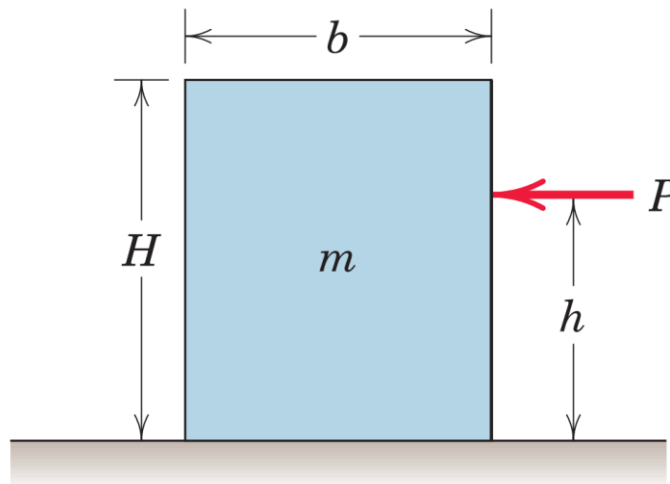
Practice at Home

5. Determine the magnitude and direction of the friction force acting on the 100-kg block shown if, first, $P = 500$ N and, second, $P = 100$ N. The coefficient of static friction is 0.20, and the coefficient of kinetic friction is 0.17. The forces are applied with the block initially at rest.



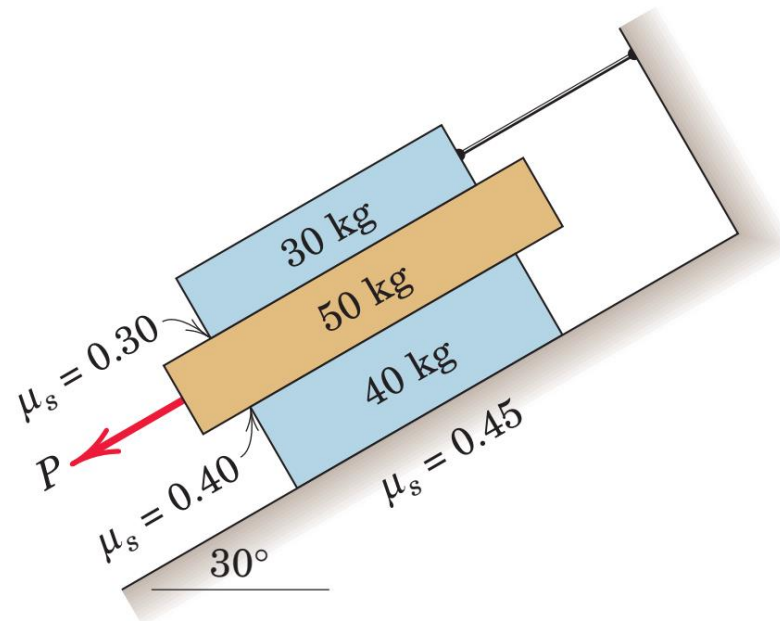
Practice at Home

6. The homogeneous rectangular block of mass m , width b , and height H is placed on the horizontal surface and subjected to a horizontal force P which moves the block along the surface with a constant velocity. The coefficient of kinetic friction between the block and the surface is k . Determine (a) the greatest value which h may have so that the block will slide without tipping over and (b) the location of a point C on the bottom face of the block through which the resultant of the friction and normal forces acts if $h = H/2$.



Practice at Home

7. The three flat blocks are positioned on the 30° incline as shown, and a force P parallel to the incline is applied to the middle block. The upper block is prevented from moving by a wire which attaches it to the fixed support. The coefficient of static friction for each of the three pairs of mating surfaces is shown. Determine the maximum value which P may have before any slipping takes place.



Practice at Home

8. The light bar is used to support the 50-kg block in its vertical guides. If the coefficient of static friction is 0.30 at the upper end of the bar and 0.40 at the lower end of the bar, find the friction force acting at each end for $x = 75$ mm. Also find the maximum value of x for which the bar will not slip.

